Introduction to the HL-LHC WP15.4 and RAC3 project

RAC3: Remote Alignment Consolidation during LS3

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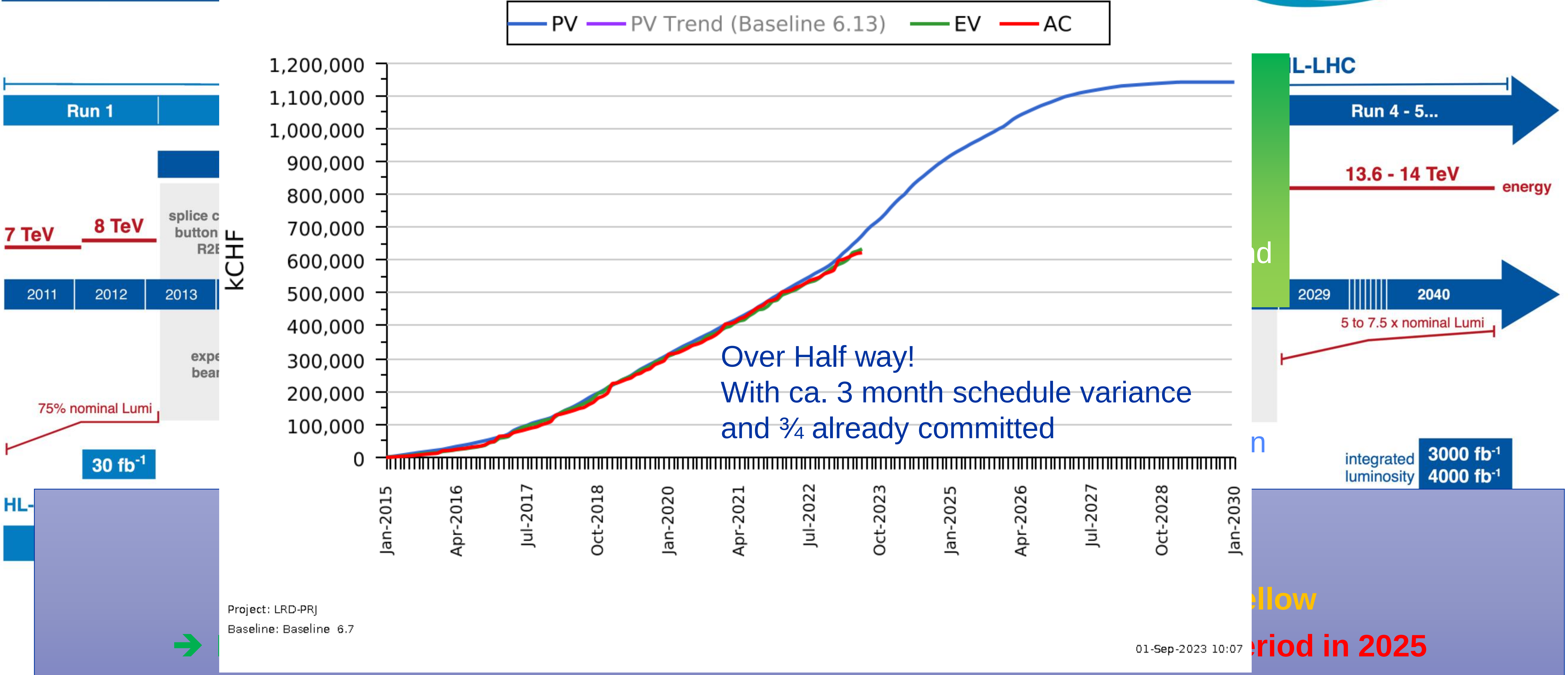
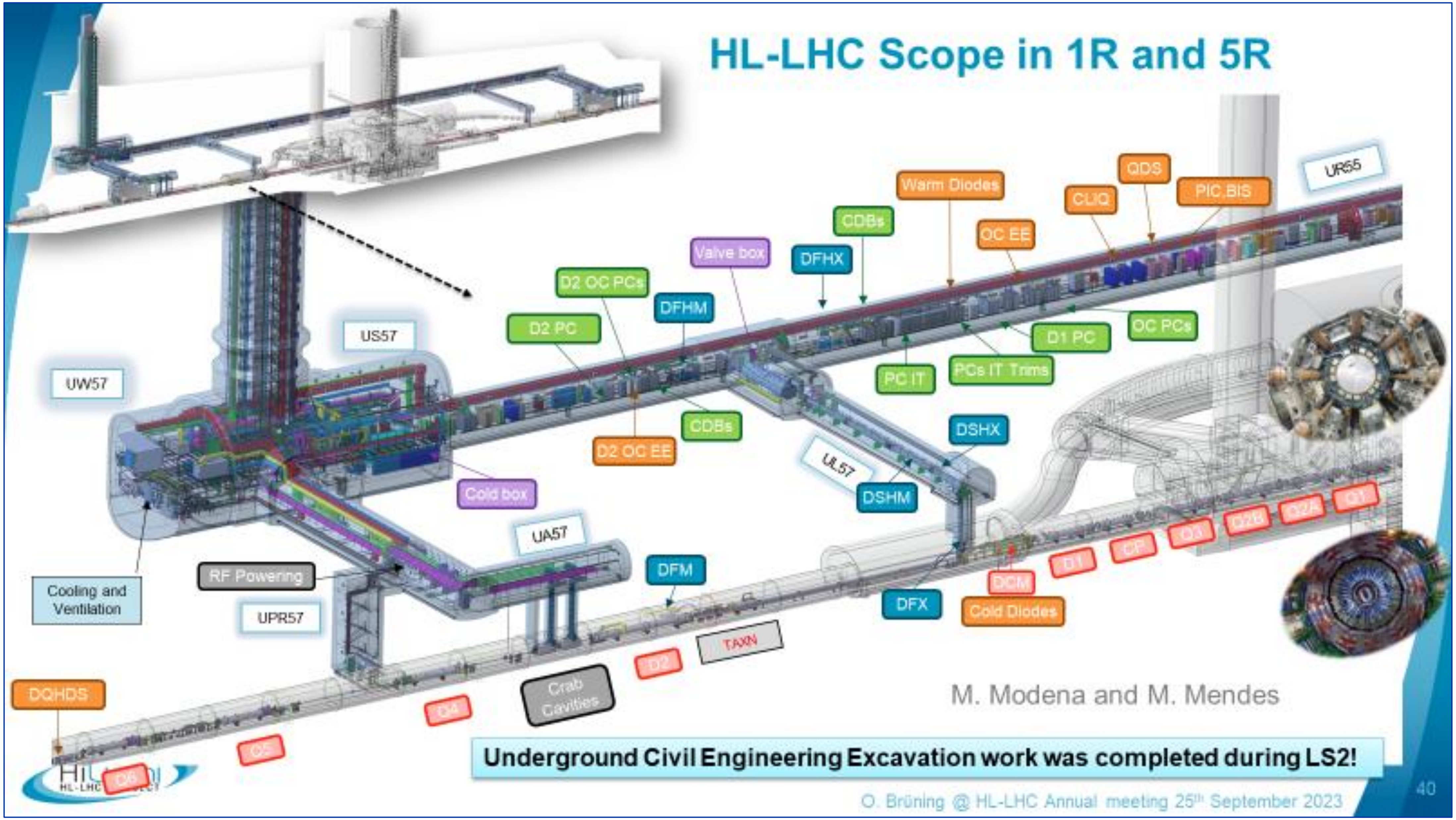
Outline

- **Introduction to HL-LHC WP15.4**
- **Introduction to RAC3 project (RAC3: Remote Alignment Consolidation during LS3)**
- **Review of the agenda**

LHC / HL-LHC Plan




HL-LHC Scope in 1R and 5R
UR55
QDS
Warm Diodes
CLIQ
PIC,BIS
CDBs
QC EE
Valve box
DFHX
D2 QC PCs
DFHM
D2 PC
US57
QC PCs
D1 PC
PCs IT Trims
PC IT
UW57
CDBs
D2 QC EE
DSHX
UL57
DSHM
Cold box
UA57
DFM
RF Powering
Cooling and Ventilation
UPR57
DFX
DCM
Cold Diodes
D1
CP
Q3
Q2B
Q2A
Q1
TAXN
D2
Crab Cavities
Q4
Q5
Q6
DQHDS
M. Modena and M. Mendes
Underground Civil Engineering Excavation work was completed during LS2!
O. Brüning @ HL-LHC Annual meeting 25th September 2023
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FRAS = Full Remote Alignment System

- It consists of alignment systems (alignment sensors, motorized adapters, their acquisition and control/command systems, associated software) allowing to determine the position of components and readjust them remotely within a range of ± 2.5 mm.
- **FRAS will provide:**
  - An important reduction of the dose taken by surveyors as no access in tunnel will be needed between YETS or LS
  - A reduction in the mechanical misalignment, allowing to decrease the required correctors strength and to push the accelerator performance
  - A gain in aperture for several components
- **All components from Q1 to Q5 will be:**
  - Either equipped with sensors and motorized axes
  - Or FRAS compatible: they are static components that can absorb the displacements of their adjacent components.
- **From the lessons learnt:**
  - In-house developments → industrialized
  - Internal monitoring of the IT cold masses and crab cavities w.r.t. their cryostats
  - Follow-up of the available stroke of the bellows

Alignment requirements for the components

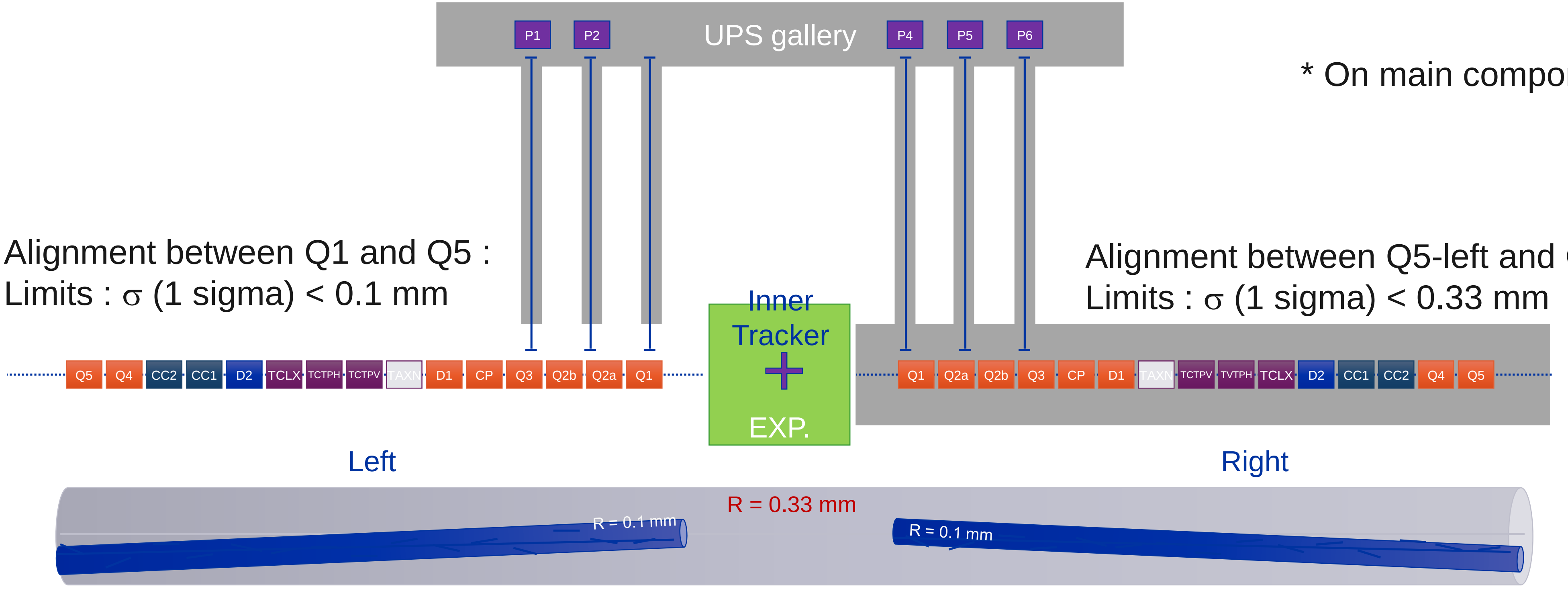
Alignment objectives (2023) for FRAS

- Position of the components cryostat along one side of the tunnel : +/- 0.1 mm *
- Position of the components cryostat along one side of the tunnel w.r.t the other side : +/- 0.33 mm *

* On main components

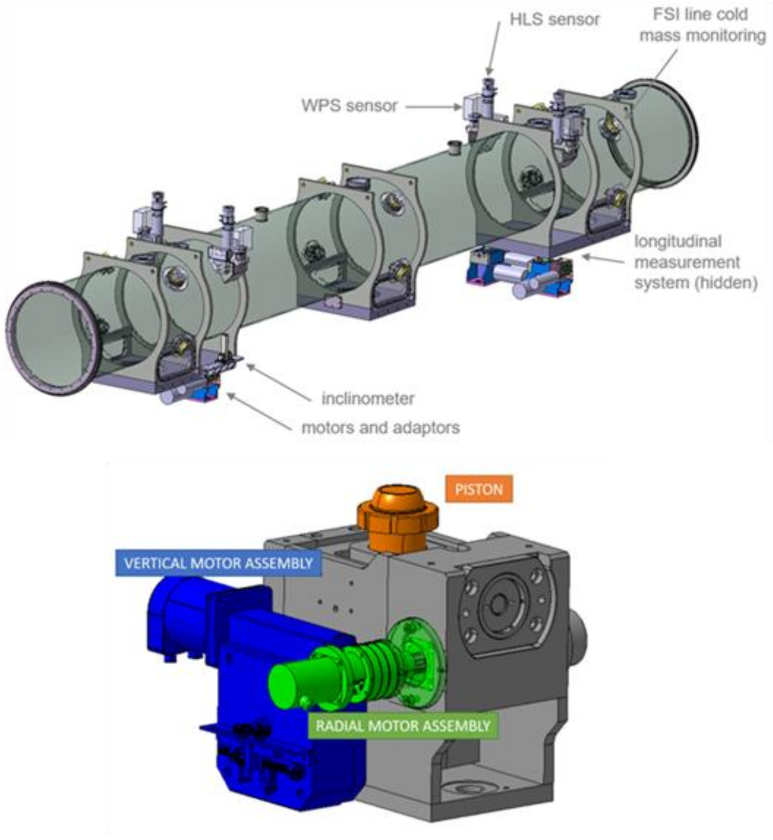
Alignment between Q1 and Q5 :
Limits : σ (1 sigma) < 0.1 mm

Alignment between Q5-left and Q5-right :
Limits : σ (1 sigma) < 0.33 mm

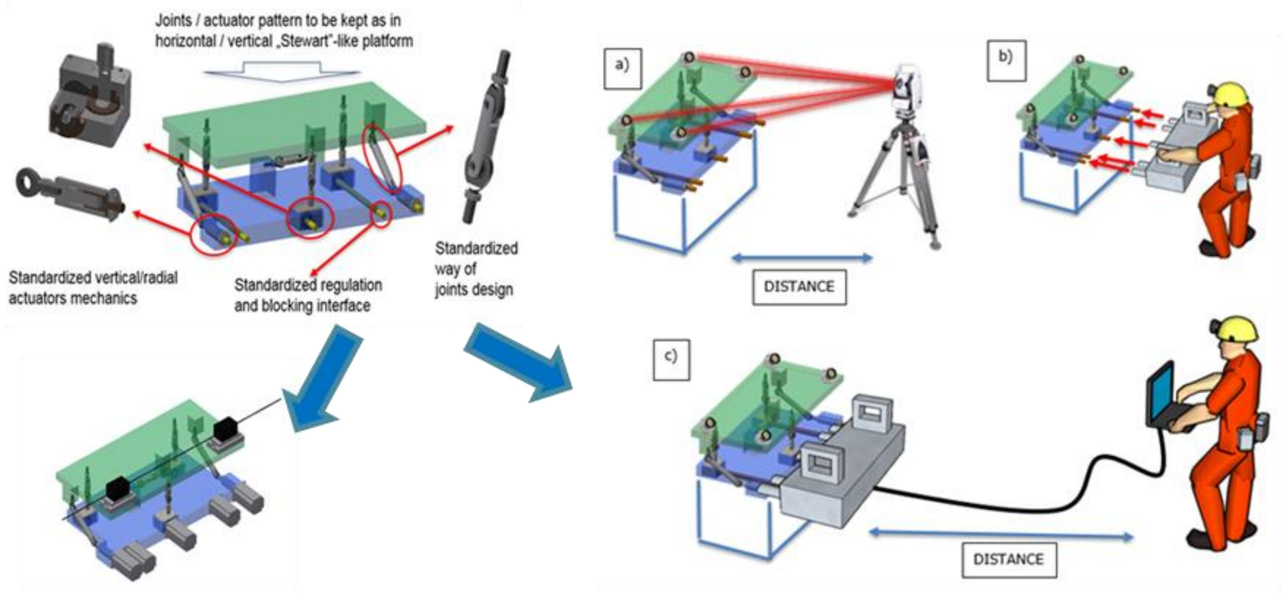


FRAS : solutions for remote adjustment

Alignment solutions for a heavy component (> 2 tons)



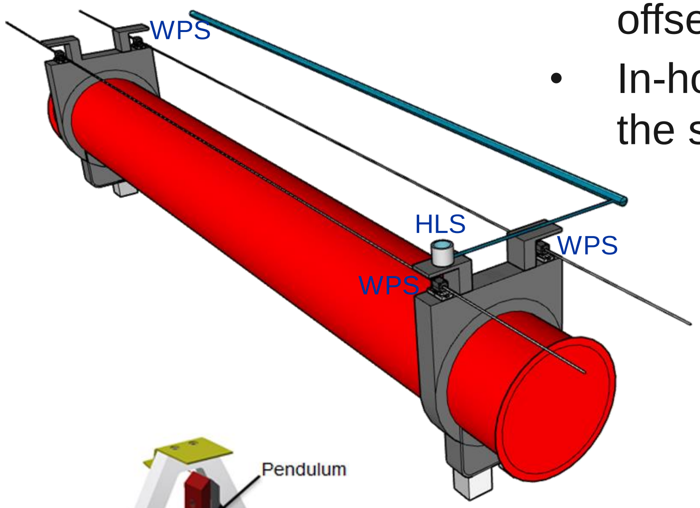
Alignment solutions for a light component (< 2 tons)



UAP platform **permanently** equipped with WPS sensors, inclinometers and motor assemblies

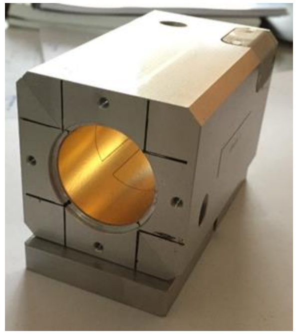
UAP platform **temporarily** equipped with plugged-in motor assemblies

FRAS: solutions for position determination



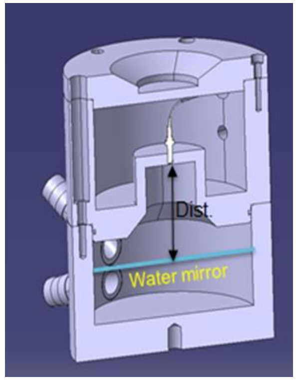
Wire Positioning Sensors (WPS):

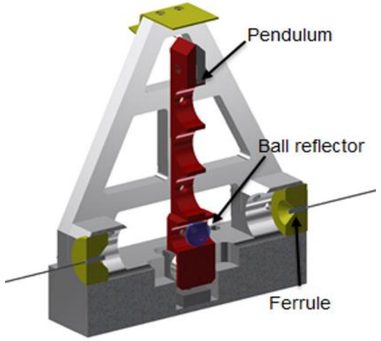
- Based on capacitive technology, performing continuous radial and vertical offset measurements w.r.t. a stretched wire, within a submicrometric resolution
- In-house design based on flexible polyimide PCB with electrodes printed on the surface and coated with gold



Hydrostatic Levelling Sensor (HLS):

- Based on the communicative vessel principle, performing vertical offset w.r.t. a water surface, within a submicrometric resolution
- Frequency Scanning Interferometry (FSI) technology
- Qualification under finalization





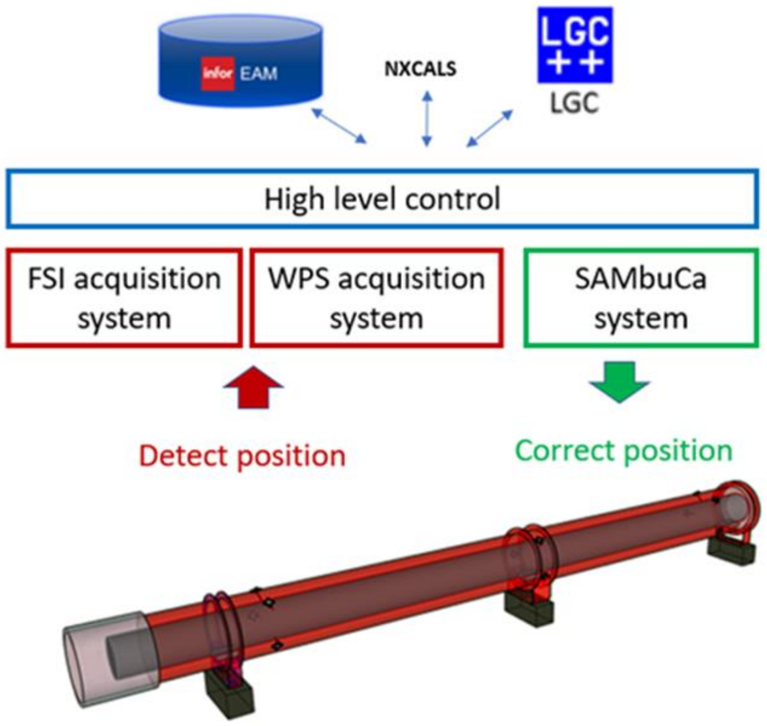
Inclinometers (in-house concept):

- Vertical pendulum measured either via capacitive or FSI technologies
- First tests: repeatability below ±10 µrad

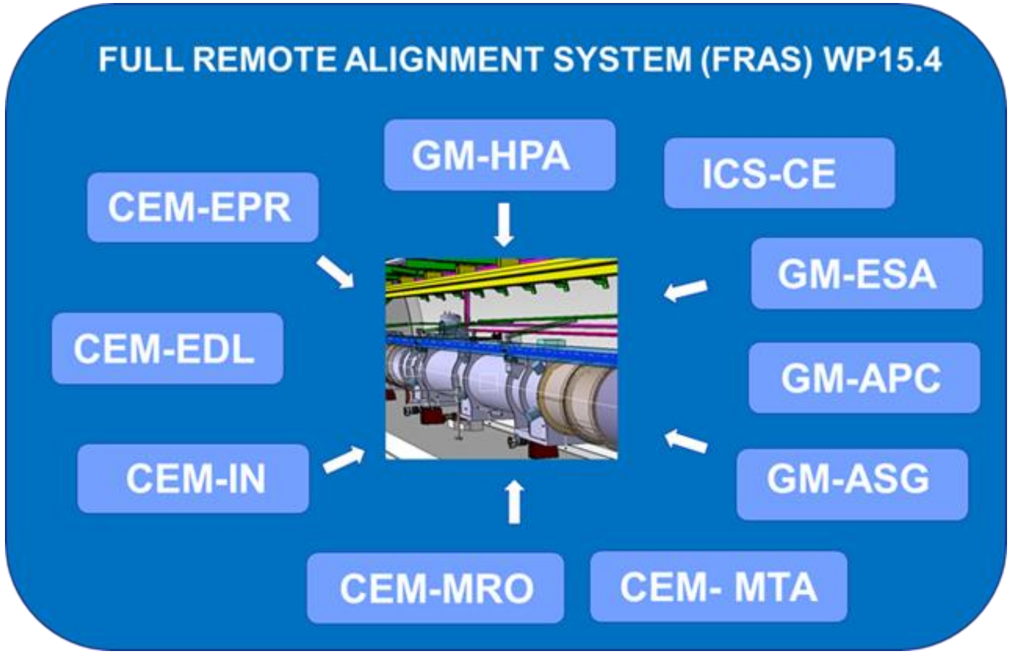
Longitudinal distance measurement

- Based on FSI technology

FRAS: control/command & acquisition aspects



Acquisition & control/command chain



68 components	1150 sensors	344 motors

FRAS qualification strategy

Development & preparation of FRAS:
- Definition of responsibilities
- Detailed definition of interfaces
- Deliverables and milestones
- Qualification of solutions on individual test setups
- Cross-comparison between alignment solutions

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FRAS qualification on 1 single component [2023]

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FRAS qualification on IT String test [2023-2025]

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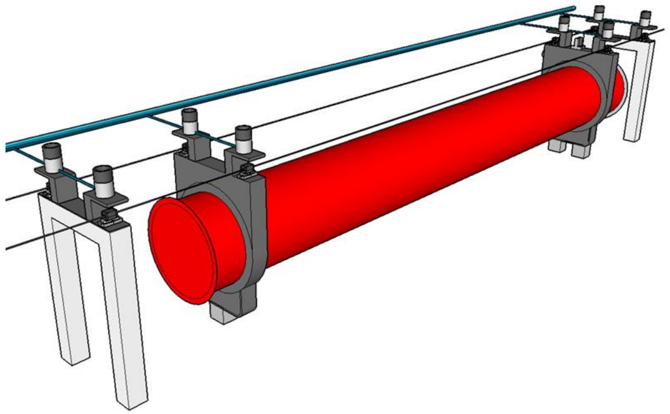
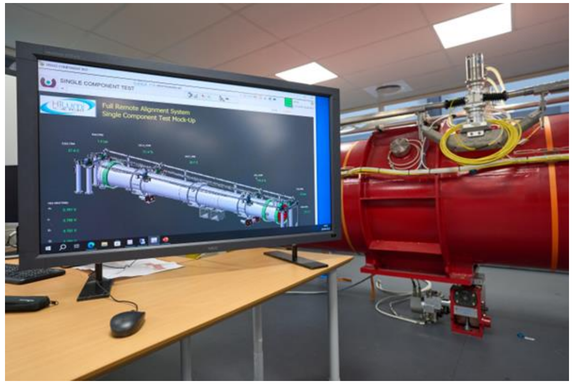
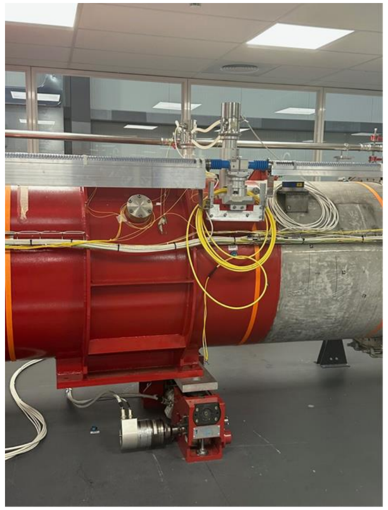
Installation, commissioning and operation in the LHC [2026+]

Single component Test:



- ✓ 8 HLS
- ✓ 8 WPS
- ✓ 2 inclinometers
- ✓ Distance sensors
- ✓ Environmental sensors
- ✓ 3 vertical adapters
- ✓ 2 radial adapters





FRAS IT String Test

IT String test objectives (for WP15.4):

- Validate the methods and procedures concerning the different alignment steps
- FRAS (*whole acquisition and control/command chain*) installation & commissioning
- Internal monitoring installation & commissioning
- Conclude on the sensors' configuration
- Qualify the internal monitoring solution & FRAS
- Validate the interlock strategy
- Check systems at their limits

Latest schedule info:

- Jacks' installation: Q2 2024
- Components installation from Q2 to Q3 2024
- **FRAS qualification: Q4 2024**
- Smoothing: Q1 2025
- 1st cool down: Q2 2025

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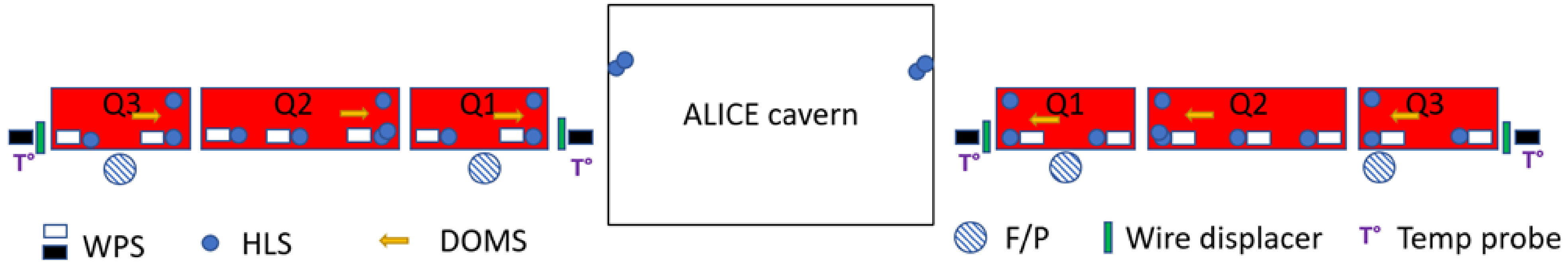
- **Introduction to HL-LHC WP15.4**
- **Introduction to RAC3 project**
- **Review of the agenda**

RAC3 in a nutshell

- Low beta triplets sensor supports, wire protection, etc. upgraded during LS2 around IP2 and IP8. sensors & motors (and associated acquisition & control/command systems) will be more than 20 years old in 2026.

- **RAC3 project: replacement of sensors and motors and their acquisition & control systems, by the same solutions than for FRAS**:
  - Replacement of current WPS sensors by **in-house capacitive WPS**, keeping the same cables transferring the signals to the remote electronics and a new sensor electronics.
  - Replacement of current HLS sensors by **FSI based HLS**, with optical fibers and FSI acquisition system.
  - Replacement of current longitudinal measurement sensors by **FSI based distance measurement sensors**, with optical fibers and FSI acquisition system
  - Replacement of the current motors by **new motors**, with a motor resolver acting as an absolute position sensor and the SAMbuCa control/command system. The mechanical interface between the jack and the motor assembly will be preserved, as the cables (the connectors will be modified).
  - The two remote diagnostics tools: filling/purging system and wire displacement system, will be kept; their acquisition and control/command system will have to be adapted.

RAC3 in a nutshell



Schedule in 3 steps:

- **EYETS 23-24: in-situ preparation** (racks, cabling modifications (WorldFIP, additional cables, AC power distribution), optical fiber installation). See ECR LHC-G-EC-0018.
- **EYETS 24-25:** exchange of **WPS** sensors + acquisition system + software part
- **Beginning of LS3:**
  - Exchange of **HLS** sensors + FSI acquisition system + software
  - Exchange of **motor assemblies** + control/command + software

Outline

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Review of the agenda

- **Objectives:**
  - Perform a **Return of Experience analysis on the SCT**: what went well, what can be improved, what has to be changed, in view of IT String Test, RAC3 and tunnel installation
  - **Extrapolate activities for the IT String Test, RAC3 and tunnel**: identify missing info, not well defined interfaces or responsibilities, issues to prepare an action plan to address them
  - This meeting will serve as **support of 2 documents under preparation**:
    - The engineering specification for the RAC3 project,
    - The engineering specification for the HL-LHC FRAS project.
- **This is an internal review between BE members**

Agenda

08:15 → 08:30 **Welcome coffee**

08:30 → 08:45 **Introduction to the FRAS and RAC3 projects, with associated milestones**

Speaker: Helene Mainaud Durand (CERN)

08:45 → 09:45 **What we want**

- 08:45 **Position calculation**
  Speaker: Vivien Rude (CERN)
- 09:00 **Adjustment strategy**
  Speaker: Mateusz Sosin (CERN)
- 09:15 **Safety strategy**
  Speaker: Borja Fernandez Adiego (CERN)
- 09:30 **Software strategy**
  Speaker: Mateusz Sosin (CERN)

09:45 → 10:45 **Motorization, diagnostics and environmental sensors**

Where we are
Next steps
Procurement, development, preparation and qualification strategy
Key interfaces

- 09:45 **Motorized adapters for jacks, UAP and other motors for diagnostic systems**
  Introduction of the motorized adapters plus motors
  Development and qualification status
  Procurement, assembly and qualification strategy
  Speaker: Michel Noir (CERN)
- 10:00 **Control/command system from CEM, including environmental sensors aquisition**
  Introduction of the control/command system from CEM
  Status
  Procurement, preparation and qualification strategy of the hardware part
  Development and qualification strategy of the software part
  Speaker: Dr Mario Di Castro (CERN)
- 10:15 **Software status from ICS**
  Introduction to the software part
  Status
  Next development steps for IT String Test, RAC3 project and LS3
  Qualification and commissioning strategy
  Speaker: Dr Enrique Blanco Vinuela (CERN)
- 10:30 **Review of interfaces**
  For the IT String Test
  For RAC3 and HL-LHC

11:15 → 12:15 **FSI based sensors and acquisition systems**

Where we are
Next steps
Procurement, development, preparation and qualification strategy
Key interfaces

- 11:15 **FSI based sensors**
  Introduction to the different sensors
  Development and qualification strategy and status for each sensor type (HLS, inclinometer, internal monitoring system, longitudinal measurements, load cells)
  Procurement strategy
  Speaker: Mateusz Sosin (CERN)
- 11:30 **FSI acquisition system**
  Introduction to the functionalities
  Development and qualification strategy and status
  DI/OT readiness
  FSI interferometer software
  Procurement, assembly and qualification strategy for IT String Test, RAC3 and HL-LHC
  Speaker: Maciej Marek Lipinski (CERN)
- 11:45 **Software part from ICS**
  Introduction to the software part to be performed by ICS
  Software development and qualification strategy and status
  Next steps and milestones for IT STring test, RAC3 and HL-LHC
  Speaker: Dr Enrique Blanco Vinuela (CERN)
- 12:00 **Review of interfaces**

12:15 → 13:00 **Capacitive sensors and their acquisition system**

Where we are
Next steps
Procurement, development, preparation and qualification strategy
Key interfaces

- 12:15 **Capacitive based alignment sensors and other types + acquisition system for WPS**
  - Introduction to capacitive based alignment sensors (and other types)
  - Presentation of the acquisition system for WPS with focus on GM part
  - Status of radiation and EMC tests
  - Development and qualification strategy and status
  - Next steps for IT String Test, RAC3 and HL-LHC project

  Speaker: Roberto Fernandez Bautista (CERN)
- 12:30 **Software part from ICS**
  Introduction to the software part from ICS
  Development and qualification strategy and status of the software part for the different sensors/acquisition systems, RAC3 and HL-LHC project.
  Speaker: Dr Enrique Blanco Vinuela (CERN)
- 12:45 **Review**

13:00 → 13:15 **Wrap-up**

13:15 → 13:45 **Sandwichs**

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